

CS-468

Software Quality Assurance & Testing

Week 10

Session 2

Statement Coverage and Test Coverage –Loops Example

WHILE LOOP

```
1  X=15
2  COUNT=0
3  WHILE X<20
4    COUNT=COUNT+1
5    X=X+1
6  END WHILE
```

```
1  X = 15
2  Count = 0
3  REPEAT
4      X = X + 1
5      Count = Count + 1
6  UNTIL X = 20
```

DO WHILE LOOP

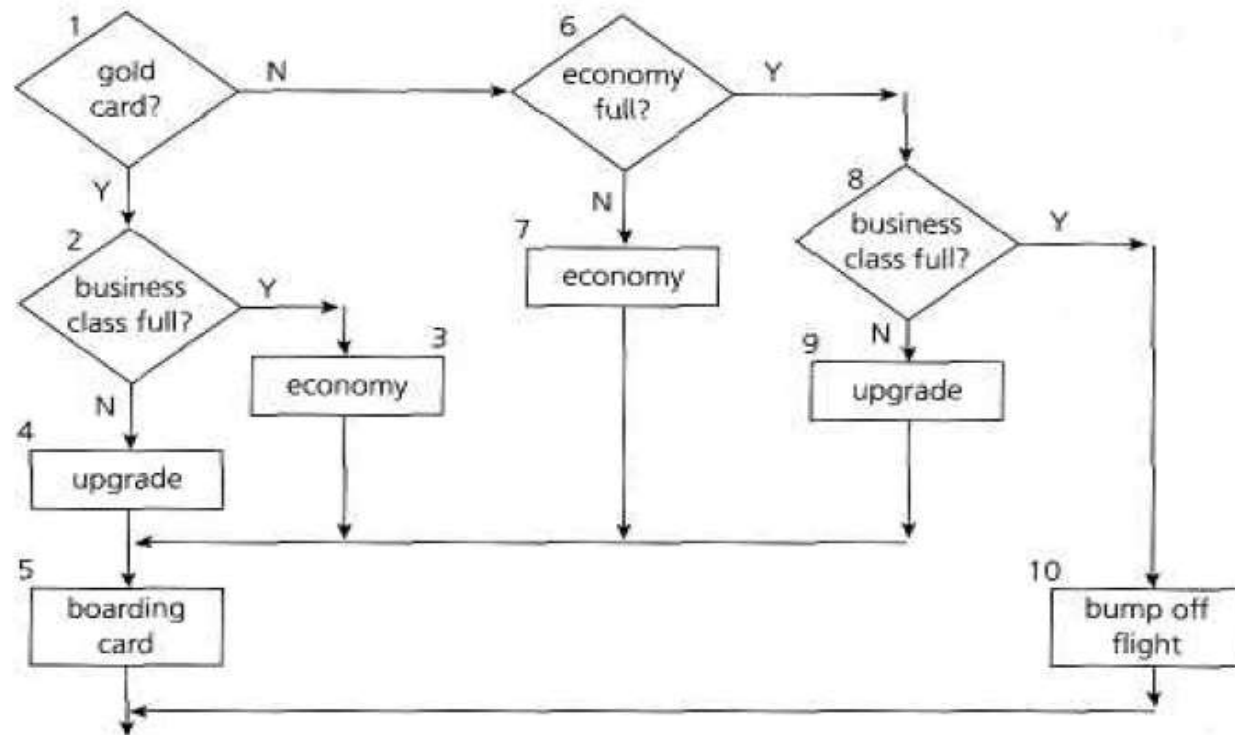
```
1  X=15
2  COUNT=0
3  DO
4    COUNT=COUNT+1
5    X=X+1
6  WHILE X<20
7  END WHILE
```

FOR LOOP

```
1  COUNT=0
2  FOR X=15 TO 20
3    COUNT=COUNT+1
4    X=X+1
5  END FOR
```

Statement Coverage and Test Coverage - Case Study

- If you are flying with an **economy ticket**, there is a possibility that you may **get upgraded to business class**, especially if you hold a **gold card** in the airline's frequent flier program. **If you don't hold a gold card, there is a possibility that you will get 'bumped' off the flight if it is full and you check in late?** This is shown in following flow chart. Note that each box (i.e. statement) has been numbered.



GENERATE TEST DATA FOR MAXIMUM DECISION COVERAGE

Statement Coverage and Test Coverage - Case Study

- Generate **test data for maximum decision coverage.**

Statement Coverage and Test Coverage - Case Study

- **Test data for maximum decision coverage** is as follows:

Card Type	Status of Business Class	Status of Economy Class	Path Exercised
Gold	Not Full	Don't Care	1, 2, 4 & 5
Gold	Full	Not Full	1, 2, 3 & 5
Not Gold	Don't Care	Not Full	1, 6, 7 & 5
Not Gold	Not Full	Full	1, 6, 8, 9 & 5
Not Gold	Full	Full	1, 6, 8 & 10

Statement Coverage and Test Coverage - Case Study

- Following three tests have been executed, what is the **statement coverage of these three tests?**
 - Test 1: Gold card holder who gets upgraded to business class
 - Test 2: Non-gold card holder who stays in economy
 - Test 3: A person who is bumped from the flight

Statement Coverage and Test Coverage - Case Study

Answer:

- Statement Coverage = Total No. of statements exercised/Total No. of statements *100
- Total No. of statements = 10

Test	Statements Exercised	Total No. of Statements Exercised	Statement Coverage (%)
1	1, 2, 4 & 5	4	$4/10 * 100 = 40\%$
2	1, 6, 7 & 5	4	$4/10 * 100 = 40\%$
3	1, 6, 8 & 10	4	$4/10 * 100 = 40\%$